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(1) Research Program/Project/Study Title: Development of <i>In-vitro</i> Mass Propagation Protocol of Four Robusta (<i>Coffea canephora</i>) Clones Through Somatic Embryogenesis	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Mambusao
Province:	Capiz
Region:	VI

(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:						
☐ Basic	☐ Pilot Testing						
☐ Applied	☐ Tech. Promotion/Commercialization						

(7) Sector/Commodity Coffee	(8) Discipline/Field of Specialization Agriculture
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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Coffee is a lucrative business and provides livelihood to many people. Luckily, the Philippines is an ideal place to grow quality coffee. Instant coffee is a popular choice among Filipinos, with an average spending of \$44 per person on instant coffee in 2022. The country is a significant player in the coffee consumption market in Asia, with an estimated 4.4% year-on-year growth rate. By 2025, the country is projected to consume 7.4 million 60-kilogram bags of coffee. Spending on coffee is forecast to grow by an average of 8.1% per year from 2022 to 2025 (<https://modernocoffee.com/philippines-coffee-consumption/>).

At present, the Philippines is only 15% self-sufficient in coffee. Mindanao produces and grows 83.63% (50,716.76 MT DC or 25,358.38 MT GCB) per year, broken down at region 10 (including Bukidnon) at 9.23% (5,601.32 MT DC or 2,800.66 MT GCB) and Region 12 (including Sultan Kudarat) at 35.60% (21,588.47 MT DC or 10,794.24 MT GCB); Luzon accounts for 9.18% while Visayas only for 7.2% (Department of Agriculture - Bureau of Agricultural Research, 2022).

Small farmers are the country's main producers of coffee in Mindanao. The four types grown are Robusta accounts for 66% of coffee produced, 25% for Arabica, 8% for Excelsa, and 1% for Liberica. However, the country's coffee supply is not enough, with coffee production supply significantly at a down trend from 72,342 MT DC (36,171 MT GCB) in 2015 to 60,640.95 MT DC (30,320.47 MT GCB) in 2020. Its yield averaged of 0.64 MT/ha 2015 declined to 0.54 MT/ha in 2020 (Department of Agriculture - Bureau of Agricultural Research, 2022).

Philippine coffee production remains relatively low and is unable to even meet local consumption demand (coffee production was at 62,062 metric tons of green coffee beans in 2019 but Filipinos consume 100,000 metric tons of coffee a year and drink almost as much coffee as the people of the United States, Brazil, Japan and the European Union) (The-Philippine-Coffee-Industry, 2020).

According to the International Coffee Organization (ICO), the global estimated number of 60-kg bags of coffee produced was 165.05 million or 9.9 million tons of GCB in 2019-2020. There was also an estimated decrease of 4.2% in global coffee production in 2019-2020 (The Philippine Coffee Industry Roadmap (2021-2025). One of the limiting factors that cause the continuous drop in coffee production is the limited access to certified planting materials.

Plant micropropagation, also known as plant tissue culture, is a technique that isolates, sterilizes, and incubates cells, tissues, or organs of chosen plants in a growth-promoting aseptic environment to create a large number of plantlets. The isolated cloning technique revealed that, given the right conditions, somatic cells may develop into a complete plant (Njukwe et al., 2007).

Traditionally, coffee is propagated through seeds and new plantations are established by planting the seedlings. However, in Robusta (*Coffea canephora*), propagation through seeds is not desirable and leads to the appearance of off type plants in the progeny due to cross pollination and segregation. Thus, *in vitro propagation* is the method of choice to palliate this problem and propagate elite mother plants in large numbers. There are several reports on micropropagation and *in vitro* multiplication of plants via somatic embryogenesis in coffee. However, successful reports on field establishment and performance of micro propagated plants of coffee are very much limited. This study will be conducted to develop a micropropagation protocol for robusta coffee to produce disease-free and certified planting materials.

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(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

The project will focus on the concept of developing micropropagation protocol of Four Robusta (*Coffea canephora*) Clones Through Direct and Indirect Somatic Embryogenesis. Traditionally the propagation of Robusta species is made through cuttings and seeds, both of these methods of propagation have disadvantages and limitations in production of clones in large number (Ismail et al., 2003). Seed propagation involves different problems such as, low germination rate, slow initial growth and short life span while vegetative cuttings guarantee uniformity seeds (Monaco et al., 1995; Ismail et al., 2003 as cited by Ameen et al., 2015).

In vitro cultivation has relevance in the clonal propagation of valuable or endangered plant germplasm, and in the production of transgenic plants (Johnson and Emino, 1979; Smith et al., 1991; Frederic et al., 2007). To date, micropropagation techniques are employed to produce a large number of new and true-to-type plants in a relatively short period. It has already proved to be successful for several members of the Rubiaceae family (Pierson et al., 1983; Carneiro and Ribeiro, 1989; Haidar, 1993; Ismail et al., 2003; Ebrahim et al., 2007; Almeasary, 2008 as cited by Ameen et al., 2015). The advantage of this technique is that it produces plants that have the same or identical characteristics as the parent, producing large numbers of seeds in a relatively short time (Asmono, et al., 2022).

Development of *In vitro* micropropagation protocol on Robusta coffee includes: 1) Evaluation of alternative sterilization technique for *in vitro* micropropagation of robusta coffee using shoot tip explants, 2) *In vitro* shoot induction and multiplication of robusta coffee supplemented with plant growth regulators (PGRs), 3) *In vitro* root induction of robusta coffee supplemented with plant growth regulators (PGRs). 4) Assessment of Genetic Fidelity of Tissue Culture-Derived Plantlets of Robusta (*Coffea canephora*) Coffee Through PCR-Based Random Amplified Polymorphic DNA (RAPD) and Inter-Simple Sequence Repeat (ISSR) Analysis, and 5) Acclimatization and Field Establishment of Tissue Culture-Derived Plantlets of Robusta (*Coffea canephora*) Coffee.

The problem in the early stages of inoculation is the contamination of the explants. Contamination can inhibit the growth process so that the explants cannot grow or die. The problem arises because the inoculated explants are not sterile. The solution to suppress contamination is to use an appropriate sterilization method. Sterilization of explants aims to prevent or kill microorganisms that may be carried or attached to the leaves when explants are taken (Asmono et al., 2022).

Somatic embryogenesis (SE) is a cloning technique based on tissue culture, where an unlimited number of identical embryos can be produced. Shoot tip explants consist of shoot apical meristem, unexpanded leaves at various development stages, and a number of leaf primordia about 1 cm in length. In shoot-tip culture the explants are inoculated in cytokinin-supplemented media. Suppression of apical dominance is caused by cytokinin and a highly branched shoot system formation is facilitated. Then manipulation of the shoot lets is done in the rooting medium to develop plantlets (Bhatia & Sharm, 2015).

Plants regenerated in tissue culture are expected to have identical genetic make-up to the mother plants and to retain intrinsic characteristics (Campos et al., 2017). One of the most crucial concerns facing *in vitro* propagation is loss of genotypic integrity of the micro propagated plants as compared to the mother plant (Rani et al., 2000).

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The objective of the study is to develop a micropropagation protocol on Robusta coffee through shoot tip cultures. Specifically, it aims to:

1. evaluate an alternative surface sterilization technique for *in vitro* micropropagation of Robusta coffee from shoot tip explants;
2. evaluate the effect of plant growth regulators, thidiazuron (TDZ), 6-benzylaminopurine (BAP) and N6-Benzyladenine (BA) at varying concentrations on shoot induction and multiplication;

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- determine the effect of indole-3-butyric acid (IBA), indole-3-acetic acid (IAA) and 1-naphthaleneacetic acid (NAA) for root initiation of Robusta coffee;
- assess the genetic fidelity of tissue culture-derived plantlets of robusta (*coffee canephora*) coffee; and
- evaluate the response of tissue culture-derived plantlets of robusta (*coffee canephora*) coffee during acclimatization and field establishment.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Importance and Botanical Description

Coffee is one of the most important crops worldwide. Coffee species grow positively between 800 to 2000 m above sea level in equatorial and tropical climates (Jaramillo, 2005). Coffee is native to Africa, and belongs to the Rubeaceae family (Farah & Dos Santos, 2015). Global production of coffee achieved 10.5 million t, while in Mexico it was 240,000 t (. International Coffee Organization, 2023). The coffee species arabica (*Coffea arabica*) and robusta (*Coffea canephora* var. Robusta) account for 70% and 30% of global production, respectively (Davis & Mannii, 2011).

It is one of the most economic important crops for more than 50 countries in the world and it is highly valuable for beverage consumption. Out of more than 100 coffee species, only *Coffea arabica* L. (Arabica type coffee) and *C. canephora* Pierre (Robusta type coffee) are commercial species (Etienne 2006). Coffee is an important beverage crop owing to its economic benefits at individual and national levels (<https://www.ncausa.org/About-Coffee/What-is-Coffee>).

Coffea canephora is the commercially most important coffee variety. Out of it, Robusta Coffee is gained. The seeds of *Coffea canephora* are very rich in caffeine. They contain up to 4% of the stimulating substance. In comparison, Arabica Coffee only contains 1-2%. Robusta Coffee plants are - like the name already tells - extraordinarily resistant and are often cultivated in West Africa and East Asia. In the Mediterranean countries, coffee mixtures with an amount of Robusta are very popular. Many Espresso lovers think that the caffeine-rich Robusta bean gives the coffee the right spice ([https://www.magicgardenseeds.com/The-Unique/Robusta-Coffee-\(Coffea-canephora\)-A.COF03-](https://www.magicgardenseeds.com/The-Unique/Robusta-Coffee-(Coffea-canephora)-A.COF03-)).

Micropropagation of Coffee

Coffee is an important source of income and employment in developing countries of Latin America, Africa and Asia (Anthony et al., 2001). According to the current context of over production and low prices of the coffee market, improvement and valorization of coffee quality could provide the coffee chain with a new impetus. In this context, the efficiency of integration of coffee quality is the main target in breeding programs as opposed to its previous status as a secondary selection criterion (Van der Vossen, 2001).

Coffee quality is a quantifiable characteristic which is related to tastes, smells, and personal preferences. Though Coffee quality is an aggregate outcome of genotype, environment, agronomic and postharvest attributes, human controlled factors largely contribute to the intrinsic quality which basically emanates due to difference in knowledge and attitudes of smallholder coffee farmers with the context of cultural managements which they were practicing in their farms (Petit, 2007). Ponte (2002) stated the importance of coffee quality rather than the quantity with respect to the producers of coffee. He argued that producers should keep in mind, the final consumers' preferences and the characteristics of the coffee for which consumers would be willing to pay more. Quality coffee provides more revenue to producers and it is a better strategy to earn more revenue for same quantity of coffee compared to low quality coffee.

The most important part of coffee in vitro micropropagation is the generation of genetically homogenous and uniform plantlets with all desirable characters of the mother plant. Somaclonal variation may lead to generation of off-types planting material with undesirable character(s) due to loss of genetic fidelity (Mayoli et al., 2019).

One method of rapid clonal propagation to produce quality coffee seeds on a large scale is tissue culture

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techniques through somatic embryogenesis. This technique can create uniform true-to-type seeds in large numbers in short time (Lizawati et al. 2023).

Genetic improvement of coffee, an important commercial crop, through classical breeding is slow and cumbersome. Biotechnology offers alternative strategies for generating new and improved coffee varieties, including those resistances to environmental extremes, pests, and diseases, low in caffeine, and with uniform fruit maturation. Large improvements in somatic embryogenesis, development of haploids, and scale-up of micropropagation have been accomplished in the last 5 years (Santana-Buzzy et al., 2007).

Importance of Surface Sterilization

Surface sterilization of explant is an important and most sensitive step in plant tissue culture. Inappropriate concentrations of sterilant have lethal effect in cell division and it restricts growth and development of explant. Both higher and lower concentrations greatly affected the cultured shoot tip explants. Lower concentrations enhance contamination whereas higher concentrations were toxic to the plant as it damaged the tissues. Sterilization is a very important aspect of tissue culture, as tissue culture aims at in vitro propagation of a desired plant material, which should be free from contamination of any other microorganism. The culture media which is prepared for the growth of explants is also conducive for the growth of other microorganisms like bacteria and fungi. Therefore, appropriate measures should be taken to avoid contamination by these microorganisms (Khan et al., 2010).

However, surface sterilization is not sufficient sometime as the internal contamination is usually not removed by this approach (Hennerty et al., 1988), especially when explants are taken from field grown plants (Savela & Uosukainen, 1994) and transferred to *in vitro* culture. There is a competition between microbes and the in vitro plants for nutrients. Mortality of the in vitro cultures increases as a result of microbial contamination and also shows the reduced growth and necrosis of cultures (Kane, 2003). In result of microbial contamination, a great loss between 3-15% is facing by most of the commercial and scientific plant tissue culture laboratories (Leifert et al., 1989).

The success of plant tissue culture protocol depends on explant sterilization (Dodds & Roberts, 1985). Selection of sterilizing agent and time period of exposure is also critical because the living material should not lose their biological activity and only contaminants should be eliminated during sterilization. Past investigations suggest that sodium hypochlorite (NaOCl) is the best choice for surface sterilization as it is readily available and can be diluted to proper concentrations. However, a balance between concentration and time must be determined empirically for each type of explant because of phytotoxicity. Ethanol (EtOH) is also used as powerful sterilizing agent but also extremely phytotoxic. Therefore, very short exposure of few minutes or seconds is generally given to explants. Surface sterilization with heavy metal salts is another way of sterilization in which Mercuric chloride (HgCl₂) is the famous one and always been a major choice but because of its toxicity and unsafe nature for both researcher and environment, it is usually replace by other disinfectants (Azam & Rahmatullah, 2009).

Plant Growth Regulators (PGRs)

Plant tissue culture, is generally dependent for its success on the inclusion of plant hormones and plant growth regulators and/or other growth active substances, as one of the five classes of required substances, in the medium. So far, a lot had been reported on the effects of different plant growth regulators (PGRs) on the *in vitro* propagation of coffee. PGRs control and modulate the initiation and development of shoots and roots on explants and embryos and also stimulate cell division and expansion (Trigiano & Gray, 2011). They also function as chemical messengers for intercellular communication.

There are currently five recognized groups of plant hormones: auxins, gibberellins, cytokinins, abscisic acid (ABA) and ethylene. They work together coordinating the growth and development of cells. The compounds that are most frequently used are called auxins, chemically called as 2, 4-dichlorophenoxy acetic acid (2, 4-D), indole-3-acetic acid (IAA), naphthalene acetic acid

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(NAA) and Indole-3-butyric acid (IBA) and cytokinins are kinetin, benzyl adenine (BA), 6-benzylaminopurine (BAP) and 2-isopentenyl aminopurine (2ip) (Chawla, 2002). Auxins, cytokinins, and auxins –cytokinins interactions are usually considered to be the most important for regulating growth and organized development in plant tissue and organ cultures, as these two classes of hormones are generally required (Thorpe, 2007).

Effect of PGRs to Callus Induction

Callus induction is the preliminary step to achieve shoot organogenesis from woody plants as explants from mature tissues fail to produce shoot buds de novo. A balance between auxin and cytokinin is most often required for the formation of adventitious shoots and roots. High levels of auxin relative to cytokinin stimulated the formation of roots, whereas high levels of cytokinin relative to auxin led to the formation of shoots (Kumar & Reddy, 2011). For example, maximum callus induction was obtained on MS medium supplemented with 2 and 10 mg/l NAA in lisianthus and gladioli, respectively (Zivyar et al., 2014), whereas the highest shoot regeneration was achieved with MS medium containing 5 mg/l BA+0.1 mg/l NAA and 4 mg/l Kin+0.5 mg/l NAA (Zivyar et al., 2014). Similar results have been obtained for callus induction of fritillary (14) and bulblet regeneration of hyacinth.

The balance of growth regulators depends on the objective of the cultivation in vitro (as e.g. shoot, root, callus or suspension culture) and on the micropropagation phase considered (initiation, multiplication or rooting) (Kumar & Reddy, 2011). Miri et al. (2011) and Mousavi et al. (2012) found that NAA was more effective than IAA and 2, 4-D on callus induction of *lisianthus* (Miri et al., 2011). For most plants, the use of auxin alone failed to induce somatic embryos. As a result, using an appropriate combination ratio of auxin and cytokinins is a key factor for somatic embryogenesis (Tokuji & Kuriyama, 2003). MS basal medium containing 1.0 mg/L 6-BA and 3.0 mg/L 2, 4-D has been the most efficient medium to display callus induction in *R. hybrida*.

Effect of PGRs to Shoot Induction and Multiplication

Result of Kahia and Owuor (1993) showed that successful in vitro multiplication on solid MS media supplemented with 8.0 mg/l BA combined with 0.8 mg/l IAA. Further, Haidar (1993) reported that the combination of 6-benzylaminopurine with 0.5 mg/l IAA produced the proliferation rate. The continuous increase in kinetin concentrations to 8.0 mg/l increased significantly the shoots number (4.4-5.2), shoot height (2.0-2.5 cm) and leaves number per ex-plant (27.4-29.2).

From the study of Ameen et al. (2015), it was evident that best clump fresh weigh was obtained from MS medium amended with BA (2.0, 4.0 and 8.0 mg/l-1) which produced 0.56, 0.64 and 0.57 g fresh weight of shoots/explant from *Coffea arabica* cvs 'Odayni', 'Burai' and 'Odayni-Bayat', respectively. The positive effect of BA on the potential to induce plant regeneration in coffee plants has been reported previously by (Kahia and Owuor, 1990; Haidar, 1993; Ebrahim et al., 2007; Almeasary, 2008) who used BA in combination with different concentrations of auxin to induce in vitro multiplication in other coffee species. Comparing the present results of this species with the previous studies, it was seen that each species needs appropriate culture medium, with appropriate concentration of growth regulator.

Effect of PGRs to Root Induction

In the previous study of Ebrahim et al. (2015), it is obvious that IAA had a significant effect on rooting of in vitro grown coffee microshoots. Increasing IAA to 3.0 mg l gave significantly the 1 highest number of roots, root length and shoot height on the three studied cultivars. No callus formation was observed on the basal part of the microshoot when 1.0 mg l IAA was used. On the 1 other hand, the 2.0 or 3.0 mg l IAA showed small 1 callus formation on the basal part of the microshoots.

This similar to previous study on *Prunus amygdalus* [27]. Better rooting in the three studied coffee cultivars was obtained when IBA auxin was used. Higher number of roots, root length and shoot height were obtained with IBA of 3.0 mg l . 1 Among the auxins tested for root initiation (i.e. IAA, IBA, NAA), IBA has been used for rooting microshoots of many Rubiaceae plants including *C. canephora*, *Lucelia*

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spp, *Rubia cordifolia*, *Gardenia Josminodies* and *Cephalis ipecacunha* [12, 13]. A recent study by Ganesh and Sreenath, [28] who reported that IBA to be suitable for the rooting of *C. arabica* and best rooting were achieved in half-strength MS medium supplemented with 1.0 mg l IBA. However, increasing NAA to 1.0 mg l gave significantly the 1 highest number of roots, root length and shoot height in comparison to all other treatments in case of Hammady and Dawaeiry cultivars. While in case of Oudayni cultivar, 2.0 mg l gave the best results. Rooting 1 parameters significantly d ecreased at 3.0 mg l NAA 1 for the three cultivars. This could be due to large callus mass (12-14 mm) around the basis of microshoots.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

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(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Study 1. Evaluation of Alternative Sterilization Technique for *In Vitro* Micropropagation of Robusta Coffee Using Shoot Tip Explants

Materials

MS medium (macroelements, microelements, iron source, vitamins), plant growth regulators (PGRs), agar, sugar, distilled water, NaOH, HCL, laminar flow cabinet, autoclave, bottles, vials, beaker, Erlenmeyer flask, graduated cylinder, pipette, analytical balance, pH meter, hot plate, shelves.

Experimental Design and Treatments

The study will be arranged in Completely Randomized Design (CRD) with three replications. There will be 10 sample explants per replication. The explants will be dipped and surface sterilized following the different treatments:

Treatments	Chemicals Used	Concentrations (mg/L/%)	Duration
1	Citric acid and Ascorbic Acid Sodium hypochlorite (NaOCL) with tween -20 Ethyl alcohol (EtOH)	200 mg/L + 100 mg/L 20% 10%	12 hours 15 minutes 10 minutes
2	Citric acid and Ascorbic Acid Sodium hypochlorite (NaOCL) with tween -20 Ethyl alcohol (EtOH)	200 mg/L + 100 mg/L 20% 10%	12 hours 15 minutes 10 minutes (twice)
3	Citric acid and Ascorbic Acid Sodium hypochlorite (NaOCL) with tween -20 Ethyl alcohol (EtOH)	200 mg/L + 100 mg/L 40% 10%	12 hours 15 minutes 10 minutes (twice)
4	Citric acid and Ascorbic Acid Sodium hypochlorite (NaOCL) with tween -20 Ethyl alcohol (EtOH) Ethyl alcohol (EtOH)	200 mg/L + 100 mg/L 40% 10% 10%	12 hours 15 minutes 20 minutes 10 minutes
5	Citric acid and Ascorbic Acid Sodium hypochlorite (NaOCL) with tween -20 Ethyl alcohol (EtOH) Ethyl alcohol (EtOH)	200 mg/L + 100 mg/L 40% 10% 10%	12 hours 15 minutes 20 minutes 20 minutes

Source of Planting Materials

Existing mother plants of four Robusta coffee clones at the Coffee Demonstration Farm - CAPSU Burias will be served as donor plants.

Explant Collection and Preparation

Shoot tips of the mother plants will be used as source of explants. 150 shoot tips will be collected that will be used in the study. Shoots (3-4 cm in length) will be collected from the donor plants. The unwanted leaves and appendages will be trimmed from the collected shoots. Trimmed shoots will be washed with soapy water, drained and placed in glass container and will be brought to the laboratory for surface sterilization treatments.

Culture Media Preparation

Standard MS (Murashige & Skoog, 1962) solid media will be used as the basal medium during the inoculation period (Table 1). For the preparation of MS media, macronutrient, micronutrient, FeEDTA, vitamins, supplemented with 3 % (w/v) sucrose and 100 mg/L Myo-inositol will be the basic medium. The

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pH of the medium will be adjusted to 5.8 using 0.1 N NaOH or 0.1 N HCL and solidified with 0.5% (w/v) agar. The medium will be dispensed at 5 ml into glass bottles, autoclaved at 15 psi for 20 minutes at 121°C and cooled.

Table 1. Nutrient composition of the Murashige and Skoog medium (1962).

CHEMICAL	SYMBOL	AMOUNT (mg/L)
<u>Macroelements</u>		
Potassium Nitrate	KNO ₃	1900
Ammonium Nitrate	NH ₄ NO ₃	1650
Calcium Chloride	CaCl ₂ ·2H ₂ O	440
Magnesium Sulfate	MgSO ₄ ·7H ₂ O	370
Potassium Phosphate	KH ₂ PO ₄	170
<u>Microelements</u>		
Manganese Sulfate	MnSO ₄ ·H ₂ O	22.3
Zinc Sulfate	ZnSO ₄ ·7H ₂ O	8.6
Boric Acid	H ₃ BO ₃	6.2
Potassium Iodide	KI	0.82
Sodium Molybdate	Na ₂ MoO ₄ ·2H ₂ O	0.25
Copper Sulfate	CuSO ₄ ·5H ₂ O	0.025
Copper Chloride	CoCl ₂ ·6H ₂ O	0.025
<u>Iron Source</u>		
Sodium EDTA	Na ₂ EDTA	37.3
Ferrous Sulfate	FeSO ₄ ·7H ₂ O	27.8
<u>Vitamins</u>		
Glycine		2.0
Nicotine acid		0.5
Pyridoxine HCL		0.5
Thiamine HCL		0.1
Myo-inositol		100.0

Inoculation and Incubation

Excision of shoot tips will be performed inside the laminar flow chamber with a pair of sterile forceps and scalpel. Shoot tips measuring 1-2 cm in length will be inoculated each unto a vial containing 5 ml of freshly prepared culture medium. Cultures will be incubated in shelves with illumination from white fluorescent tubes and a room temperature of 25±2°C.

The percent tissue survival will be recorded after four weeks of culturing date.

Data to be Gathered

- Days to contamination.** This will be determined by counting the days from inoculation to the day where visible signs of contamination will be noted on the cultured explant after two weeks.
- Percent contamination (%).** This will be determined by counting the number of cultures showing signs of contamination and expressing it as percentage of the total number of cultures per replicate.
- Types of contaminants.** Contaminants will be determined whether fungi or bacteria based on visible manifestations.
- Percent tissue survival (%).** This will be determined by counting the number of cultures that shows signs of greening, without tissue browning and contamination and will be expressed as percentage of the total number of cultures per replicate.

Statistical Tools and Analysis

Data will be statistically analyzed using the computer software Statistical Tool for Agricultural Research (STAR) version 2.0.1 2014. The presence of significant differences among treatments will be determined using Analysis of Variance (ANOVA) in CRD. Comparison of treatment means will be done using Least Significant Difference (LSD) Test at 5% level of significance.

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Study 2. In vitro shoot induction and multiplication of robusta coffee supplemented with plant growth regulators (PGRs)

Materials

MS medium (macroelements, microelements, iron source, vitamins), plant growth regulators (PGRs), agar, sugar, distilled water, NaOH, HCL, laminar flow cabinet, autoclave, bottles, vials, beaker, Erlenmeyer flask, graduated cylinder, pipette, analytical balance, pH meter, hot plate, shelves.

Experimental Design and Treatments

The study will be arranged in Completely Randomized Design (CRD) with three replications. There will be 10 sample explants per replication. The treatment will be the following:

- T₁ – MS alone (control)
- T₂ – MS + 2.0 mg/l BA (N6-Benzyladenine)
- T₃ – MS + 4.0 mg/l BA
- T₄ – MS + 6.0 mg/l BA
- T₅ – MS + 8.0 mg/l BA
- T₆ – MS + 2.0 mg/l Kinetin (6-furfurylaminopurine)
- T₇ – MS + 4.0 mg/l Kinetin
- T₈ – MS + 6.0 mg/l Kinetin
- T₉ – MS + 8.0 mg/l Kinetin

Source of Planting Materials

Existing mother plants of four Robusta coffee clones at the Coffee Demonstration Farm - CAPSU Burias will be served as donor plants.

Explant Collection and Preparation

Shoot tips of the mother plants will be used as source of explants. 270 shoot tips will be collected that will be used in the study. Shoots (3-4 cm in length) will be collected from the donor plants. The unwanted leaves and appendages will be trimmed from the collected shoots. Trimmed shoots will be washed with soapy water, drained and placed in glass container and will be brought to the laboratory for surface sterilization treatments.

Culture Media Preparation

Standard MS (Murashige & Skoog, 1962) solid media will be used as the basal medium during the inoculation period (Table 1). One week after inoculation, the cultures will then be transferred to full strength of MS media supplemented with various plant growth regulators (PGRs). For the preparation of MS media, macronutrient, micronutrient, FeEDTA, vitamins, fortified with 3 % (w/v) sucrose and 100 mg/L Myo-inositol will be the basic medium. PGRs will be added before autoclaving. The pH of the medium will be adjusted to 5.8 using 0.1 N NaOH or 0.1 N HCL and solidified with 0.7% (w/v) agar. The medium will be dispensed at 5 ml into glass bottles, autoclaved at 15 psi for 20 minutes at 121°C and cooled.

Inoculation and Incubation

Excision of shoot tips will be performed inside the laminar flow chamber with a pair of sterile forceps and scalpel. Shoot tips measuring 1-2 cm in length will be inoculated each unto a vial containing 5 ml of freshly prepared culture medium for shoot induction and multiplication. Cultures will be incubated in shelves with illumination from white fluorescent tubes and a room temperature of 25±2°C.

The cultures will be observed for four (4) weeks and then will be transferred to the same medium (fresh medium) for every four (4) weeks. Data will be collected 60 days after inoculation (DAI).

Data to be Gathered

1. **Days to shoot formation.** This will be determined by counting the days from inoculation to the day where visible signs of shoots will be noted.

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$$2. \text{ Shoot Induction Percentage (\%)} = \frac{\text{No. of survived explants}}{\text{No. of explants cultured}} \times 100$$

3. **Number of shoots/cultures.** This will be determined by counting the number of shoots per culture.
4. **Shoot length/culture (cm).** This will be determined by measuring the length of the shoot per culture.
5. **Number of internodes.** This will be determined by counting the number of internodes per culture.
6. **Number of nodes.** This will be determined by counting the number of nodes per culture.
7. **Number of leaves/cultures.** This will be determined by counting the number of leaves per culture.

Statistical Analysis

Data will be statistically analyzed using the computer software Statistical Tool for Agricultural Research (STAR) version 2.0.1 2014. The presence of significant differences among treatments will be determined using Analysis of Variance (ANOVA) in CRD. Comparison of treatment means will be done using Duncan's Multiple Range Test (DMRT) Test at 5% level of significance.

Study 3. In vitro root induction of robusta coffee supplemented with plant growth regulators (PGRs)

Materials

MS medium (macroelements, microelements, iron source, vitamins), plant growth regulators (PGRs), agar, sugar, distilled water, NaOH, HCL, laminar flow cabinet, autoclave, bottles, vials, beaker, Erlenmeyer flask, graduated cylinder, pipette, analytical balance, pH meter, hot plate, shelves.

Experimental Design and Treatments

The study will be arranged in a Completely Randomized Design (CRD) with three replications. Each replication will be composed of 10 samples. Two weeks after, the shoots will be then be separated individually and will be placed at the treatments designated below:

- T₁ – MS alone
- T₂ – ½ strength MS + 1.0 mg/L IBA (Indole-3-butyric acid)
- T₃ – ½ strength MS + 2.0 mg/L IBA
- T₄ – ½ strength MS + 3.0 mg/L IBA
- T₅ – ½ strength MS + 1.0 mg/L IAA (Indole-3-acetic acid)
- T₆ – ½ strength MS + 2.0 mg/L IAA
- T₇ – ½ strength MS + 3.0 mg/L IAA

Plant Materials

Shoots derived from the multiplication step will be used in this study. These will be cultured on medium without hormones for two weeks to eliminate the effects of hormones applied in the multiplication stage.

Inoculation and Incubation

Micropropagated plantlets will be inoculated each unto a vial containing 5 ml of freshly prepared culture medium for root induction. Cultures will be incubated in shelves with illumination from white fluorescent tubes and a room temperature of 25±2°C.

Data to be Gathered

1. **Number of days to root formation.** This will be determined by counting the number of days from inoculation to root formation.

$$\frac{\text{Number of plants producing roots}}{\text{Total number of plants}} \times 100$$

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2. **Rooting percentage (%)** = _____

3. **Number of roots per shoot.** This will be determined by counting the number of shoots per shoot.

4. **Root length (cm).** This will be determined by measuring the length of the roots.

Statistical Analysis

Data will be statistically analyzed using the computer software Statistical Tool for Agricultural Research (STAR) version 2.0.1 2014. The presence of significant differences among treatments will be determined using Analysis of Variance (ANOVA) in CRD. Comparison of treatment means will be done using Duncan's Multiple Range Test (DMRT) Test at 5% level of significance.

Study 4. Assessment of Genetic Fidelity of Tissue Culture-Derived Plantlets of Robusta (*Coffea canephora*) Coffee Through PCR-Based Random Amplified Polymorphic DNA (RAPD) and Inter-Simple Sequence Repeat (ISSR) Analysis

Study 5. Acclimatization and Field Establishment of Tissue Culture-Derived Plantlets of Robusta (*Coffea canephora*) Coffee

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Year 1				Year 2				Year 3			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
1. Surface sterilization												
2. Shoot induction and multiplication												
3. Root induction												
4. Assessment of Genetic Fidelity												
5. Acclimatization												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

- Patent/Intellectual Property-** surface sterilization technique, shoot induction and multiplication, root induction processes, 3 IECs
- People Services-** trainings/demo with potential coffee farmers
- Partnership-** Funding Agencies/ Coffee Farmer's Association

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(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities) 1. Surface sterilization technique 2. Shoot induction and multiplication 3. Root induction			
(17) Target Beneficiaries (Who the clienteles are and what are the expected outcome/effects of the use of the projects output) 1. Potential coffee farmers/growers 2. Coffee Farmer's Association			

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(18) Personnel Requirement			
Name	Role (Program, Project, Study Leader)	Field of Specialization	
Percent time devoted to the Program/ Project/ Study			
Program/Project Leader (in hours):			
Program/Project Members (in hours):			
(19) Budgetary Requirement: Php 300, 000.00 (Repair and Maintenance, Refrigerator, bottles, vials, some chemicals)			
SOURCES OF EXPENSES		Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)			
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)			
- Transportation allowance			
- Meal allowance during data gathering			
- Professional fees (laboratory analysis and other related fees)			
- Miscellaneous (for incidental expenses)			
- Printing			
C. CAPITAL OUTLAY			
Total			
(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, http://researchjournal.capsu.edu.ph/)			
Ameen, A., AL-Azab, Sanad, Habib, S. M., Mohammed, A., Hussein & El-Sherif, F. (2015). Micropropagation of Four Coffee Cultivars (<i>Coffea arabica</i> L.) from Yemen through Shoot Tip Culture. Hortscience Journal of Suez Canal University, (4): 25-31. Anthony, F., Bertrand, B., Quiros, O., Lashermes, P., Berthaud, J., & Charrier, A. (2001). Genetic diversity of wild coffee (coffea arabica l.) Using molecular markers. Euphytica, 118: 53-65. Asmono, S.L., Wardana, R., & Rahmawati. (2022). Optimization of the sterilization method for leaf explant Robusta BP 308 coffee in vitro. IOP Conf. Series: Earth and Environmental Science 980 (2022) 012001, doi:10.1088/1755-1315/980/1/012001 Azam, F. M. S. & Rahmatullah, M. (2009). Tissue culture of a Year-Round Fruiting Variety of <i>Artocarpus heterophyllus</i> L. in Bangladesh. <i>Acta Hort.</i> 806 Bhatia, S., Sharma, K., & Bera T. (2015). Modern Applications of Plant Biotechnology in Pharmaceutical Sciences. Pharmaceutical Sciences. http://doi.org/10.1016/C2014-0-02123-5 Campos, N.A., Panis, B., & Carpentier, S.C. (2017), "Somatic embryogenesis in coffee: The evolution of biotechnology and the integration of omics technologies offer great opportunities", <i>Frontiers in Plant Science</i>, 8 (1460) Davis, A.P. 2011. <i>Psilanthus Mannii</i>, the Type Species of <i>Psilanthus</i>, Transferred to <i>Coffea</i>. <i>Nord. J. Bot.</i>, 29, 471–472. [CrossRef] Department of Agriculture - Bureau of Agricultural Research, 2022 Dodds, J. H., & Roberts, L. W. (1985). <i>Experiments in Plant Tissue Culture</i>. 2nd Edn., Cambridge Univ. Press. Cambridge, UK., pp: 232 Etienne, H. 2006. Somatic embryogenesis protocol: Coffee (<i>Coffea arabica</i> L. and <i>C. canephora</i> P.). In: Protocol for somatic embryogenesis in woody plants. Jain S.M., Guta P.K. (eds). Springer. The Netherlands, p. 167-179 Farah, A., & Dos Santos, T.F. (2015). <i>The Coffee Plant and Beans: An Introduction</i>; Elsevier Inc.: Amsterdam, The Netherlands, 2015; ISBN 9780124167162. Available online: https://www.researchgate.net/profile/Adriana_Farah2/publication/267967496_The_coffee_plant_and_beans_Introduction_Chapter_1/links/546541710cf2f5eb17ff3901/The-coffee-			

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plant-and-beans-IntroductionChapter-1.pdf Haidar, A. K. A. (1993). Studies on the Marjoram (<i>Origanum vulgare</i> L.) and Syrian micropropagation of <i>Coffea arabica</i> L. - M.Sc. marjoram (<i>Origanum syriacum</i> L.). M.Sc. thesis, Thesis, Zagazig University, Egypt Hennerty, M. E., Upton, M. E., Furlong, P.A., James, D.J., Harris, D.P., & Eaton, R. A. (1988). Microbial contamination of in vitro cultures of apple rootstocks M26 and M9. <i>Acta Hort.</i>, 225: 129-138 International Coffee Organization. World Coffee Production. Available online: https://www.ico.org/prices/po-production.pdf (accessed on 23 February 2023). Jaramillo, R.A. (2005). La Agroclimatología Del Cafeto. In <i>Clima Andino y Café en Colombia</i>; CENICAFE: Manizales, Colombia, 149–157. Available online: https://biblioteca.cenicafe.org/bitstream/10778/859/1/Portada.pdf Kahia, J.W. (1993). Plantlet regeneration from <i>Coffea arabica</i> L. shoot tips. <i>Kenya-Coffee</i>, 58: 1467-1471. Kane, M. 2003. Bacterial and fungal indexing of tissue cultures. http://plant-tc.cfans.umn.edu/listserv/1996/log9612/indexing.htm Khan, R., Zerega, N., Hossain, S., & Zuberi, M. I. (2010). Jackfruit (<i>Artocarpus heterophyllus</i> Lam.) diversity in Bangladesh: Land use and artificial selection. <i>Economic Botany</i>, 64 (2):124-136 Kumar, N. & Reddy, M. P. (2011). <i>In vitro</i> plant propagation: a review. <i>Journal of Forest Science</i> 27(2):61-72 Kumar, V., Madhava, N., & Ravishankar, G.A. (2006). Developments in coffee biotechnology-in vitro plant propagation and crop improvement. <i>Plant Cell Tiss. Org. Cult.</i> 87: 49-65. Leifert, C., Waites, W. M., & Nicholas, J. R. (1989). Bacterial contaminants of micropropagated plant cultures. <i>J. Applied Microbiol.</i> 67: 353-361 Lizawati, L., Zulkarnain, Z., Antony, D., & Purnamaningsih, R. (2022). The effect of 2,4-D, BA and Thidiazuron on somatic embryo induction of liberica coffee of Tungkal Composite from Jambi. <i>ISEPROLOCAL</i>. https://doi.org/10.1051/e3sconf/202337303012 ISE Mayoli, R.N., Isutsa, D.K. & Nyende, Ab.B. (2019). “A protocol for regenerating genetically stable coffee (<i>Coffea arabica</i> L.) plantlets through direct somatic embryogenesis”, <i>International Journal of Development and Sustainability</i>, Vol. 8 No. 9, pp. 645-656. Miri, S. M., Savari, A., & Behzad, K. (2011). Effect of auxins on callogenesis in lisianthus. 7th International Symposium on in Vitro Culture and Horticultural Breeding, Ghent, Belgium Mousavi, E. S., Behbehani, M., Hadavi, E. & Miri, S. M. (2012). Callus induction and plant regeneration in lisianthus (<i>Eustoma granditlorum</i>). <i>Trakia Journal of Sciences</i> 10 (1): 22-25 Muniswamy, B., Kosaraju, B., Mishra, M.K. & Yenugula, R. (2017). Field Performance and Genetic Fidelity of Micropropagated Plants of <i>Coffea canephora</i> (Pierre ex A. Froehner). <i>Open Life Sci.</i>, 12: 1–11. DOI 10.1515/biol-2017-0001 Ponte, S. (2022). “The ‘Latte Revolution’? Regulation, Markets and Consumption in the Global Coffee Chain” <i>World Development</i>. 30(7);1099-1122. Rani, V., Singh K.P., Shiran, B., Nandy, S., Devaramath, R.M., Sreenath, H.L., & Raina, S.N. (2000), “Evidence for new nuclear and mitochondrial genome organizations among high-frequency somatic embryogenesis derived plants of allotetraploid <i>Coffea arabica</i> L. (Rubiaceae)”, <i>Plant Cell Reports</i>, Vol. 19 No. 10, pp. 1013- 1020. Santana-Buzzy, N., Rojas-Herrera, R., & Galaz-Ávalos, R.M. (2007). Advances in coffee tissue culture and its practical applications. In <i>Vitro Cell.Dev.Biol.-Plant</i> 43, 507–520. https://doi.org/10.1007/s11627-007-9074-1 Savelle, M. L. & Uosukainen, M. (1994). Characterization of bacteria contaminating tissue cultures of apple rootstock. <i>J. Applied Bacte.</i>, 76: 368-376 Van der Vossen, H. A. M. (2001). Agronomy I: Coffee Breeding practices. In: Clarke RJ, Vitzthum OG (eds), <i>Coffee: recent developments</i>, pp.184-201. Blackwell Science, Oxford, Great Britain. Zivyar, S., Miri, S. M. & Rahimi, M. A. (2014). The effect of plant growth regulator of BAP and NAA on callus formation from corm explants of <i>Gladiolus</i>. 1st National Ornamental Congress. The-Philippine-Coffee-Industry, 2020 The Philippine Coffee Industry Roadmap (2021-2025) https://www.da.gov.ph/wp-content/uploads/2019/06/Philippine-Coffee-Industry-Roadmap-2017-2022.pdf				

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(21) Capsule Curriculum Vitae (One-page brief C.V. only)			